Awni Altabaa

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Research Interests

Novel Neural Network Architectures · *Theoretical Foundations of Modern ML Methods* · Out-of-Distribution Generalization · Inductive Biases · Compositional & Relational Reasoning in Neural Networks · Reinforcement Learning · Information-Theoretic Perspectives on Learning and Reasoning · Large Language Models · LLM Agents · Connections Between Cognitive Neuroscience and Machine Intelligence

Education

2022 – 2027 (expected)	◇ **Ph.D. in Statistics & Data Science.** Yale University — New Haven, Connecticut.
2022 – 2024	◇ **Master's in Statistics.** Yale University — New Haven, Connecticut.
2018 – 2022	◇ **Bachelor's in Applied Mathematics.** Queen's University — Kingston, Ontario.

Relevant Graduate Coursework: Advanced Optimization Techniques, Measure-Theoretic Probability, Decision Theory, Theory of RL, High-dimensional Statistics, Statistical Inference on High-Dim Graph-Structured Data, Harmonic Analysis on Graphs, Statistical Learning Theory, Sampling & Optimal Transport

Overview of Recent Research

Developing architectures with enhanced relational reasoning and systematic compositional generalization.

- In our ***ICLR*** paper [o], we propose a novel neural module called the *Abstractor* for learning relational representations within the Transformer framework, motivated by an architectural inductive bias for relational learning that disentangles relational information from object-level features.
- Based on the ideas developed in the Abstractor [o], our ***TICS*** paper [o] with collaborators in cognitive neuroscience proposes the concept of a *relational information bottleneck* as a possible mechanisms for abstraction in the human mind and brain.
- In our ***TMLR*** paper [o], we propose a neural architecture based on a convolution-like operation that captures higher-order hierarchical relational patterns among groupings of objects.
- In our follow-up ***ICML*** work [o], we introduce the *Dual-Attention Transformer (DAT)*, a novel architecture based on Transformers that integrates sensory information (object properties) and relational information (relationships between objects) with distinct attention mechanisms.
  See the project webpage for more details: www.awni.xyz/dual-attention.
  Open Source Implementation: www.pypi.org/project/dual-attention.

Theoretical Characterization of Statistical Aspects of Modern Practice in Artificial Intelligence.

- In our ***NeurIPS*** spotlight paper [o], we develop a statistical theory of learning under chain-of-thought (CoT) supervision, introducing an information-theoretic measure to quantify the additional discriminative power gained from observing the reasoning process, and demonstrate that CoT supervision can yield significantly faster learning rates compared to standard "end-to-end" supervision.
  See the project webpage for more details: www.awni.xyz/cot-info.
- In our ***NeurIPS*** paper [o], we study how the structure of causal dependence between system variables affects the statistical complexity of reinforcement learning in general sequential decision-making problems, providing a more powerful model of partial observability phenomena.

Research Publications

1. **A. Altabaa**, S. Chen, J. Lafferty, and Z. Yang, "Unlocking out-of-distribution generalization in transformers via recursive latent space reasoning," 2025. arXiv: 2510.14095 [cs.LG], Project webpage: www.awni.xyz/algorithmic-generalization-transformer-architectures.

2. **A. Altabaa**, O. Montasser, and J. Lafferty, “CoT Information: Improved Sample Complexity under Chain-of-Thought Supervision,” *Neural Information Processing Systems* (***NeurIPS***), *spotlight*, 2025, Project webpage: www.awni.xyz/cot-info.

3. **A. Altabaa** and J. Lafferty, “Disentangling and integrating relational and sensory information in transformer architectures,” *International Conference on Machine Learning* ***(ICML)***, 2025, Project webpage: www.awni.xyz/dual-attention.

4. **A. Altabaa** and Z. Yang, “On the role of information structure in reinforcement learning for partially-observable sequential teams and games,” *Neural Information Processing Systems* ***(NeurIPS)***, 2024.

5. **A. Altabaa**, T. W. Webb, J. D. Cohen, and J. Lafferty, “Abstractors and relational cross-attention: An inductive bias for explicit relational reasoning in transformers,” in *International Conference on Learning Representations* ***(ICLR)***, 2024, Project webpage: www.awni.xyz/abstractor.

6. **A. Altabaa** and J. Lafferty, “Learning hierarchical relational representations through relational convolutions,” *Transactions on Machine Learning Research* ***(TMLR)***, 2024, Project webpage: www.awni.xyz/relational-convolutions.

7. T. W. Webb, S. M. Frankland, **A. Altabaa**, S. Segert, K. Krishnamurthy, D. Campbell, J. Russin, T. Giallanza, R. O’Reilly, J. Lafferty, *et al.*, “The relational bottleneck as an inductive bias for efficient abstraction,” *Trends in Cognitive Sciences* ***(TICS)***, 2024.

8. **A. Altabaa** and J. Lafferty, “Approximation of relation functions and attention mechanisms,” *Information Theory, Probability and Statistical Learning: A Festschrift in Honor of Andrew Barron*, 2024.

9. **A. Altabaa**, B. Yongacoglu, and S. Yüksel, “Decentralized multi-agent reinforcement learning for continuous-space stochastic games,” in *2023 American Control Conference* ***(ACC)***, 2023.

10. **A. Altabaa**, D. Huang, C. Byles-Ho, H. Khatib, F. Sosa, and T. Hu, “Genedragnn: Gene disease prioritization using graph neural networks,” in *IEEE Conference on Computational Intelligence in Bioinformatics and Computational Biology* ***(CIBCB)***, 2022.

Industry Experience

Summer 2025 ◇ **Quantitative Researcher Intern.** Citadel Securities.
Alpha research @ Systematic Equities / FICC. Project: LLM agent + Alpha Modeling.

Technical Skills

Python · Deep Learning Frameworks (PyTorch, Tensorflow, & JAX) · LLM Agent Frameworks · Version Control (Git) · High-Performance Computing (SLURM, Multi-GPU/TPU) · Efficient Distributed Training: DDP/FSDP/DeepSpeed, Mixed Precision, Profiling, Checkpointing · Language Model Pretraining, Fine-Tuning, and Evaluation · Reproducible ML: Config Mgmt, Seeding/Determinism, Unit Tests · Experiment Tracking (W&B, MLflow) · Data Visualization and Analysis (Matplotlib, Seaborn, Plotly)

Misc

Awards & Honors ◇ NSF AI Institute for Artificial and Natural Intelligence Fellow/Trainee

Review Service ◇ Journal of Data Science, NeurIPS 2024, ICLR 2025, NeurIPS 2025